

Genetic report for longevity

Appetite, Satiety and Eating Behaviour

Leptin-melanocortin (MC4R, FTO), incretins and response to agonists GLP-1/GIP, taste (sweet, bitter, fat), reward and food preferences

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Most of the adiposity genes act on the brain and move on appetite and satiety, not on metabolism: MC4R, FTO, TMEM18, NEGR1, SH2B1, BDNF. This report reads these genes, the incretin receptors (GLP1R, GIPR), which are the targets of the retatrutide that already takes, the taste (TAS2R38, TAS1R2, CD36, FGF21) and the reward (DRD2/ANKK1), and polygenic scores of the size of the portions, the variety of the diet and food preferences. Everything is about eating behavior and adiposity.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

23

variants

17

in the chip

4

imputed
($DR2 \geq 0.8$)

2

not found

9

Scores with
percentile

Overall reading. The control of appetite is favourable. In the homeostatic signal: MC4R without risk alleles in the two main signals (rs17782313 T/T and rs12970134 G/G), SH2B1 A/A and GNPDA2 A/A in favor, and BDNF Val66Met (1 copy of Met, in favor). FTO rs9939609 T/A, NEGR1 G/A and TMEM18 C/C (the common genotype) are typical. The Health Report puts the tendency to eat in excess at the 26th percentile and snack in the 22 ("less likely"); portion size in the 44. In preferences: FGF21 G/G (no allele preference for sweet), processed meat at the 10th percentile and avoid sugar in the 93, both in favor. The 20 score loci of sweetness is at the 87th percentile, against; both signals contradict each other. I like very palatable foods in 35 and fruit in 49. TAS2R38 heterozygous (sensitive to bitterness of the crucifers). In reward: DRD2/ANKK1 G/G (without the A1 allele) and CNR1 C/T, without judgment. The point to be monitored is the anorexia nervosa score at the 95th percentile (9 loci) and 1 allele of the CADM1: it is not diagnostic, but together with the retatrutide points to eat less, not more. GLP1R (Ser168/Ser168, Leu7/Leu7) and GIPR (Glu354/Glu354) are informative for the response to retatrutide.

Favourable

- **Never eat sugar (2816 variants):** the 93rd percentile
- **Processed meat (2346 variants):** the 10th percentile
- **Snacking (6 variant score):** the 22nd percentile (Health Report)
- **MC4R rs17782313 TT:** MC4R: C allele reduces satiety signaling and increases ingestion, especially of snacks.
- **MC4R rs12970134 GG:** MC4R: second common sign; allele A increases adiposity.
- **SH2B1 rs7498665 AA:** SH2B1: amplifies leptin and insulin signaling.
- **GNPDA2 rs10938397 AA:** GNPDA2: expressed in hypothalamus.
- **BDNF rs6265 TC:** BDNF Val66Met: Val allele (C in the strand read) is the most adiposity in GWAS; BDNF stops appetite downstream of MC4R.
- **FGF21 rs838133 GG:** FGF21: Allele A is associated with more sweet and carbohydrates and less protein in the diet.

Unfavourable

- **Neurotic anorexia:** the 95th percentile
- **I like sweets:** the 87th percentile
- **CADM1 rs6589488 TA:** CADM1: locus of anorexia nervosa.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 17 chip genotypes, 0 needed complement, 2 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Homeostatic sign: leptin and melanocortins

Leptin informs the hypothalamus of the reserves; the POMC-MC4R pathway stops appetite. FTO (rs9939609) affects ghrelin and preference for dense foods. The effects of each variant are small (about 1–3% in ingestion).

Ageing mechanisms: Deregulation of Nutritional Sensing; Endocrine dysregulation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Size of portions (591 variants)	Typical / intermediate	the 44th percentile · z=-0.25	Higher score : larger portions. PGS001518 (PGS Catalog; snpnet; Portion size; PGP000244, PMID 35324888). Variants with genotype: 497 of 591 (covering of 84.2% coefficients).
Trend to eat in excess (8 variant score)	Typical / intermediate	the 26th percentile	It labels "typical likelihood". Source: Health Report.
Snacking (6 variant score)	Favourable	the 22nd percentile	It labels "less likely". Source: Health Report.

Favourable

- Snacking (6 variant score) (the 22nd percentile)
- MC4R rs17782313 TT
- MC4R rs12970134 GG
- SH2B1 rs7498665 AA
- GNPDA2 rs10938397 AA
- BDNF rs6265 TC

Unfavourable

None.

Favourable variants (5)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MC4R	rs17782313 18:60183864 · T/A/C	TT	chip	C (0; 24%)	strong	MC4R: C allele reduces satiety signaling and increases ingestion, especially of snacks. T/ T: 0 risk alleles (matched with the Health Report).
MC4R	rs12970134 18:60217517 · G/A	GG	chip	A (0; 27%)	strong	MC4R: second common sign; allele A increases adiposity.
SH2B1	rs7498665 16:28871920 · A/G/T	AA	chip	G (0; 33%)	moderate	SH2B1: amplifies leptin and insulin signaling.
GNPDA2	rs10938397 4:45180510 · A/G	AA	chip	G (0; 42%)	moderate	GNPDA2: expressed in the hypothalamus.
BDNF	rs6265 11:27658369 · C/T	TC	chip	C (1; 80%)	moderate	BDNF Val66Met: Val allele (C in the strand read) is the most adiposity in GWAS; BDNF stops appetite downstream of MC4R.

Typical / intermediate variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MC4R	rs2229616 18:60372043 · C/T	CC	chip	C (2; 99%)	moderate	MC4R V103I: the Ile allele (T in the strand read) protects; C/C is the common one.
FTO		TA	chip	A (1; 41%)	strong	

	rs9939609 16:53786615 · T/A					FTO: Allele A is associated with more ghrelin after meals, less satiety and preference for energy-dense foods.
TMEM18	rs6548238 2:634905 · T/ C/G	CC	chip	C (2; 83%)	strong	TMEM18: expressed in the hypothalamus; allele C increases adiposity.
NEGR1	rs2815752 1:72346757 · G/A/C	GA	chip	A (1; 64%)	moderate	NEGR1: neuronal growth in the hypothalamus.

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
LEPR	rs1137101 1:65592830 · A/ G/T	AG	chip	—	weak	LEPR Gln223Arg: inconsistent associations. Information.
LEP	rs7799039 7:128238730 · G/A/C	AG	imputation DR2=1.00	—	weak	LEP -2548G>A: leptin levels; small effect. Information.

Incretins and response to GLP-1/GIP agonists

GLP1R and GIPR are the retatrutide receptors (with the glucagon receptor). Variants of these receptors are associated, in small studies, with more or less response to agonists; none changes the treatment decision.

Ageing mechanisms: Insulin resistance; Intercellular Communication

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
GLP1R	rs6923761 6:39066296 · G/A/C	AA	imputation DR2=1.00	—	weak	GLP1R Gly168Ser: the Ser (A) allele was associated with more fat loss and more effect of GLP-1 agonists in some small studies. Information. A/A = Ser168/Ser168.
GLP1R	rs10305420 6:39048860 · C/T	TT	imputation DR2=0.98	—	weak	GLP1R Pro7Leu: T allele was associated with a different response to agonists in small studies. Information. T/T = Leu7/Leu7.
GIPR	rs1800437 19:45678134 · G/C	GG	chip	—	moderate	GIPR Glu354Gln: the Gln allele (C in the strand read) alters the GIP signaling and is associated with less adiposity in GWAS. Information for the GIP component of the retatrutide. G/G = Glu354/Glu354 (the common genotype).

Palate and preferences

TAS2R38 (cruciferous plume), TAS1R2 (sweet), CD36 (fat) and FGF21 (sweet and hydrate preference, against protein). The preferences scores come from questionnaires of the UK Biobank.

Ageing mechanisms: Deregulation of Nutritional Sensing; Dysbiosis

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Never eat sugar (2816 variants)	Favourable	the 93rd percentile · z=1.44	Higher score : more tendency to avoid sugar. PGS000991 (PGS Catalog; snpnet; Never eat sucking; PGP000244, PMID 35324888). Variants with genotype: 2472 of 2815 (covering of 88.2% coefficients).
Fresh fruit (4257 variants)	Typical / intermediate	the 49th percentile · z=0.03	Higher score = more fruit. PGS001062 (PGS Catalog; snpnet; Fresh fruit intake; PGP000244, PMID 35324888). Variants with genotype: 3774 of 4257 (covering of 89.9% coefficients).
Processed meat (2346 variants)	Favourable	the 10th percentile · z=-1.26	Higher score : more processed meat. PGS001067 (PGS Catalog; snpnet; Processed meat intake; PGP000244, PMID 35324888). Variants with genotype: 2050 of 2346 (covering of 87.6% coefficients).
I like sweets	Unfavourable	the 87th percentile · z=1.07	Higher score : more taste for sweets. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 17 of 20 loci with genotype. Studies: Graham CA (Sweet food like); May-Wilson S (F-sweet food like) May-Wilson S (Milky sweets like). Variants with genotype: 17 of 20 (covering of 86.1% coefficients).
I like very palatable foods	Typical / intermediate	the 35th percentile · z=-0.40	Higher score : more taste for fatty, salty and sweet foods. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 21 of 22 loci with genotype. Studies: May-Wilson S (F-fatty/salty food like) May-Wilson S (F-highly palatable foods like)). Variants with genotype: 21 of 22 (covering of 95.2% coefficients).
Fat proportion in the diet	No judgement (informative)	the 66th percentile · z=0.41	Higher score = more fat in proportion of energy. Information. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 8 of 9 loci with genotype. Studies: Merino J (Related fat intake). Variants with genotype: 8 of 9 (covering of 66.2% coefficients).

Favourable

- Never eat sugar (2816 variants) (the 93rd percentile)
- Processed meat (2346 variants) (the 10th percentile)
- FGF21 rs838133 GG

Unfavourable

- I like sweets (the 87th percentile)

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FGF21	rs838133 19:48756272 · A/C/G/T	GG	chip	A (0; 42%)	moderate	FGF21: Allele A is associated with more sweet and carbohydrates and less protein in the diet.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TAS2R38	rs713598 7:141973545 · C/A/G/T	CG	chip	—	moderate	TAS2R38 A49P: with rs1726866 and rs10246939 forms the haplotypes PAV (sensitive to bitterness) and AVI (unsensitive). Information. Heterozygous; with the other two SNP,

						haplotypes PAV/AVI (sensitive to bitterness, according to Health Report).
TAS1R2	rs35874116 1:18854899 · T/C	CC	chip	—	weak	TAS1R2 Ile191Val: Val allele was associated with less sugar consumption in people with high adiposity. Information.
CD36	rs1761667 7:80615623 · G/A	GG	chip	—	weak	CD36 -31118G>A: the A allele lowers the sensitivity to the flavor of fat. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FTO	— — · —	—	SNP not found in the data	—	moderate	FTO: main variant of preferences for sweets and very palatable foods. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Reward and eating behavior

Dopamine (DRD2/ANKK1) and endocannabinoids (CNR1) affect food reward; anorexia nervosa has its own genetics, shared with metabolic and psychiatric traits.

Ageing mechanisms: Intercellular communication alteration; Neuroinflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Diet variability (1353 variants)	No judgement (informative)	the 98th percentile · z=2.01	Question about varying the diet from week to week. Information. PGS001068 (PGS Catalog; snpnet; Variation in diet; PGP000244, PMID 35324888). Variants with genotype: 1180 of 1353 (covering of 87.6% coefficients).
Neurotic anorexia	Unfavourable	the 95th percentile · z=1.64	Higher score : higher risk. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 8 of 9 loci with genotype. Studies: Duncan L (Anorexia nervosa); Watson HJ (Anorexia nervosa). Variants with genotype: 8 of 9 (covering of 81.0% coefficients).

Favourable

None.

Unfavourable

- Anorexia nervosa (the 95th percentile)
- CADM1 rs6589488 TA

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CADM1	rs6589488 11:115226236 · A/C/G/T	TA	imputation DR2=0.99	A (1; 14%)	moderate	CADM1: locus of anorexia nervosa. Meaning: GWAS Catalog: Anorexia nervosa (Watson HJ 2019; p=6e-11); allele A ↑

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
DRD2/ANKK1	rs1800497 11:113400106 · G/A	GG	chip	—	weak	Taq1A: the A1 allele lowers the density of D2 receptors in the striatum; associations with inconsistent food reward. Information. G/G = A2/A2, without the allele A1.
CNR1	rs1049353 6:88143916 · C/A/T	CT	chip	—	weak	CNR1 1359G>A: small associations with adiposity and appetite. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NCAM1	— · —	—	SNP not found in the data	—	moderate	NCAM1: locus of anorexia nervosa. No major variant of GWAS Catalog for this gene with usable genotype and sense.

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Intercellular Communication	Anorexia nervosa (the 95th percentile); CADM1 TA	—	—
Neuroinflammation	Anorexia nervosa (the 95th percentile); CADM1 TA	—	—
Nutrient Sensing Deregulation	I like sweets (the 87th percentile)	Snacking (6 variant score) (the 22nd percentile); MC4R TT; MC4R GG; SH2B1 AA; GNPDA2 AA; BDNF TC; Never eat sugar (2816 variants) (the 93rd percentile); Processed meat (2346 variants) (the 10th percentile); FGF21 GG	Size of portions (591 variants) (the 44th percentile); Tendency to eat in excess (8 variant score) (the 26th percentile); MC4R CC; FTO TA; TMEM18 CC; NEGR1 GA; Fresh fruit (4257 variants) (the 49th percentile); I like very palatable foods (the 35th percentile)
Dysbiosis	I like sweets (the 87th percentile)	Never eat sugar (2816 variants) (the 93rd percentile); Processed meat (2346 variants) (the 10th percentile); FGF21 GG	Fresh fruit (4257 variants) (the 49th percentile); I like very palatable foods (the 35th percentile)
Endocrine dysregulation	—	Snacking (6 variant score) (the 22nd percentile); MC4R TT; MC4R GG; SH2B1 AA; GNPDA2 AA; BDNF TC	Size of portions (591 variants) (the 44th percentile); Tendency to eat in excess (8 variant score) (the 26th percentile); MC4R CC; FTO TA; TMEM18 CC; NEGR1 GA
Insulin resistance	—	—	—

Practical part

Ideas for appetite, crossed with the genetic plan v2 and current medication (isotretinoin, retatrutide, dapagliflozin, telmisartan, tadalafil, rapamycin). The current stack remains as it is; everything is suggestion (not applied). With retatrutide and this genetics, the risk is to eat little protein and little fiber, not to eat too much.

1. Protein first, ~140 g/day in 3-4 meals

Genetics: MC4R without risk alleles; p95 anorexia score

With low appetite, starting each meal by protein (30-40 g) guarantees the daily dose and protects lean mass. Do not skip meals for lack of hunger.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

2. More fiber and vegetables at the beginning of the meal

Genetics: I like p87 sweets (loci score)

Legumes and protein before hydrates lower the glucose peak and increase satiety. More fiber is better: legumes, oats, seeds, whole fruit.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

3. Cooked crucifers

Genetics: TAS2R38 heterozygous (sensitive to bitterness)

Cooking or jumping broccoli and cabbages, with olive oil and lemon, takes part of the bitter; facilitates eating them every day.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

4. Sweets away from home, whole fruit instead

Genetics: I like p87 sweets; avoid p93 sugar

The genetic signs of sweetness contradict each other; the environment decides more. Having whole fruit by hand and not keeping sweets at home.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

Avoiding or with caution

- Jump meals or make long fasts with the retatrutide (less protein, loss of muscle). *suggestion (not applied)*
- To replace meals with sweet shakes or ultra-processed bars. *suggestion (not applied)*
- Eating quickly or in front of screens (satiety arrives late). *suggestion (not applied)*

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found. The rare mutations of MC4R, leptin and POMC (monogenic obesity) are not read by chip.
- The variants of response to GLP-1/GIP agonists come from small studies, especially with liraglutide and exenatide; there are no data with retatrutide.
- With the retatrutide, the current appetite is mainly pharmacological; the genes describe the base trend, which again weighs if the drug stops.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001518 (PGS Catalog; snpnet; Portion size; PGP000244, PMID 35324888 · PGS000991 (PGS Catalog; snpnet; Never eat sucking; PGP000244, PMID 35324888 · PGS001062 (PGS Catalog; snpnet; Fresh fruit intake; PGP000244, PMID 35324888 · PGS001067 (PGS Catalog; snpnet; Processed meat intake; PGP000244, PMID 35324888 · PGS001068 (PGS Catalog; snpnet; Variation in diet; PGP000244, PMID 35324888).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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- Tanigawa Y et al. *PLoS Genet* 2022 (PGS000991, PGS001062, PGS001067, PGS001068, PGS001518).
- GWAS Catalog and PGS Catalog (2026); Ensembl REST (public alleles and frequencies).

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